

Computational Fluid Dynamics for Aerospace Applications

Lecture 10

Compressible Flows: Part II

Compressible Flows: Part II

- Isentropic Nozzles
- Area-Mach Number Relationships
- Converging-Diverging Nozzles
- Friction Losses in Nozzles

Isentropic Nozzles

- Recall our stagnation properties

- $h_t = h + \frac{1}{2} V^2$

- $T_t = T \left(1 + \frac{\gamma-1}{2} M^2 \right)$

- $p_t = p \left(1 + \frac{\gamma-1}{2} M^2 \right)^{\frac{\gamma}{\gamma-1}}$

- $\rho_t = \rho \left(1 + \frac{\gamma-1}{2} M^2 \right)^{\frac{1}{\gamma-1}}$

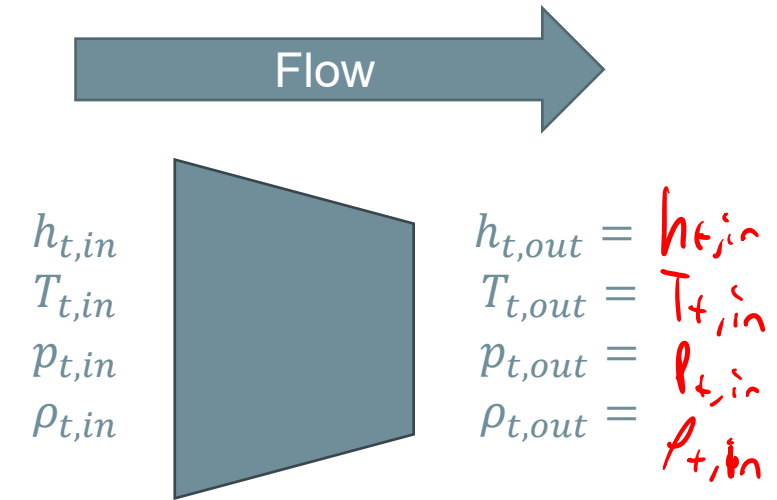
- These are **constant** for an isentropic process



$$dh_t = 0$$

Isentropic Nozzles

- Consider an isentropic nozzle
 - How do stagnation properties vary?
- Enthalpy
 - $h_{t,in} = h_{t,out}$
 - $h_{in} + \frac{1}{2}V_{in}^2 = h_{out} + \frac{1}{2}V_{out}^2$
- Other variables
 - $T_{t,in}/T_{t,out} = 1$
 - $p_{t,in}/p_{t,out} = 1$
 - $\rho_{t,in}/\rho_{t,out} = 1$



Isentropic Nozzles

$$T_{+1} = T_{+2}$$

$$T_+ = T \left(1 + \frac{\gamma-1}{2} M^2 \right)$$

• State changes

$$\bullet \frac{T_2}{T_1} = \frac{1 + \frac{\gamma-1}{2} M_1^2}{1 + \frac{\gamma-1}{2} M_2^2}$$

$$\bullet \frac{p_2}{p_1} = \left(\frac{1 + \frac{\gamma-1}{2} M_1^2}{1 + \frac{\gamma-1}{2} M_2^2} \right)^{\frac{\gamma}{\gamma-1}}$$

$$\bullet \frac{\rho_2}{\rho_1} = \left(\frac{1 + \frac{\gamma-1}{2} M_1^2}{1 + \frac{\gamma-1}{2} M_2^2} \right)^{\frac{1}{\gamma-1}}$$

• Area change

$$\bullet \frac{A_2}{A_1} = \frac{M_1}{M_2} \left(\frac{1 + \frac{\gamma-1}{2} M_2^2}{1 + \frac{\gamma-1}{2} M_1^2} \right)^{\frac{\gamma+1}{2(\gamma-1)}}$$

+ mass conservation

In the isentropic case, all changes are related to the change in Mach number

$$\rho_1 V_1 A_1 = \rho_2 V_2 A_2$$

Isentropic Nozzles

• Example

- Consider an isentropic converging nozzle with air

- $M_1 = 0.4$

- $T_1 = 300 \text{ K}$

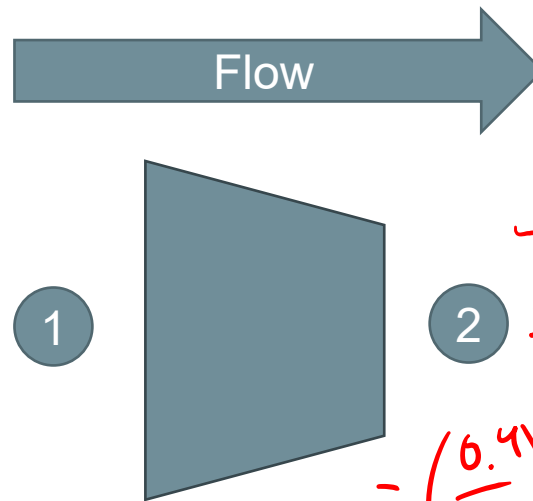
- $p_1 = 101325 \text{ Pa}$

- $\gamma = 1.4$

- $R = 287 \frac{\text{J}}{\text{kg-K}}$

- $M_2 = 0.8$

- Compute $T_2, p_2, \rho_2, V_2, A_2/A_1$



$$T_2 = T_1 \left(\frac{1 + \frac{\gamma-1}{2} M_1^2}{1 + \frac{\gamma-1}{2} M_2^2} \right) = (300 \text{ K}) \left(\frac{1 + \frac{1.4-1}{2} (0.4^2)}{1 + \frac{1.4-1}{2} (0.8^2)} \right) = (300 \text{ K}) \left(\frac{1 + 0.12}{1 + 0.288} \right) = (300 \text{ K}) \left(\frac{1.12}{1.288} \right) = 274.47 \text{ K}$$

$$p_2 = p_1 \left(\frac{1 + \frac{\gamma-1}{2} M_1^2}{1 + \frac{\gamma-1}{2} M_2^2} \right)^{\frac{\gamma}{\gamma-1}} = (101325 \text{ Pa}) \left(\frac{1 + \frac{1.4-1}{2} (0.4^2)}{1 + \frac{1.4-1}{2} (0.8^2)} \right)^{\frac{1.4}{1.4-1}} = (101325 \text{ Pa}) \left(\frac{1.12}{1.288} \right)^{10} = 74,218 \text{ Pa}$$

$$\rho_2 = \frac{p_2}{RT_2} = \frac{(74,218 \text{ Pa})}{(287 \frac{\text{J}}{\text{kg-K}})(274.47 \text{ K})} = 0.942 \frac{\text{kg}}{\text{m}^3}$$

$$M = V/a \Rightarrow V = aM \Rightarrow V_2 = M_2 \sqrt{\gamma RT_2}$$

$$V_2 = (0.8) \sqrt{(1.4)(287 \frac{\text{J}}{\text{kg-K}})(274.47 \text{ K})} = 265.67 \text{ m/s}$$

$$\frac{A_2}{A_1} = \frac{M_1}{M_2} \left(\frac{1 + \frac{\gamma-1}{2} M_2^2}{1 + \frac{\gamma-1}{2} M_1^2} \right)^{\frac{\gamma+1}{2(\gamma-1)}} = \left(\frac{0.4}{0.8} \right) \left(\frac{1 + \frac{1.4-1}{2} (0.8^2)}{1 + \frac{1.4-1}{2} (0.4^2)} \right)^{\frac{1.4+1}{2(1.4-1)}} = 0.38 = \frac{A_2}{A_1}$$

For isentropic system, $a = \sqrt{\left. \frac{dp}{d\rho} \right|_s} \Rightarrow d\rho = \frac{1}{a^2} dp$

Area-Mach Number Relationships

- Recall our “one-dimensional” flows

Momentum:

$$dp + \rho V dV = 0$$

Mass:

$$\frac{d\rho}{\rho} + \frac{dA}{A} + \frac{dV}{V} = 0$$

$$dV = - \left(\frac{dA}{A} + \frac{d\rho}{\rho} \right) V$$

Combine mass/momentum

$$dp = \rho V^2 \left(\frac{dA}{A} + \frac{d\rho}{\rho} \right)$$

$$dp = \rho V^2 \left(\frac{dA}{A} + \frac{1}{a^2} \frac{1}{\rho} dp \right)$$

$$dp = \rho V^2 \frac{dA}{A} + \frac{V^2}{a^2} dp$$

$$dp - M^2 dp = \rho V^2 \frac{dA}{A}$$

$$\bullet \frac{dp}{dA} = \left(\frac{\gamma M^2}{1 - M^2} \right) \frac{p}{A}$$

$$dp(1 - M^2) = \rho V^2 \frac{dA}{A}$$

$$dp(1 - M^2) = \rho \gamma \frac{V^2}{a^2} \frac{dA}{A}$$

$$dp(1 - M^2) = \rho \gamma M^2 \frac{dA}{A}$$

$$\frac{dp}{dA} = \frac{\gamma M^2}{1 - M^2} \frac{p}{A}$$

for ideal gas,

$$a = \sqrt{\frac{\gamma p}{\rho}}$$

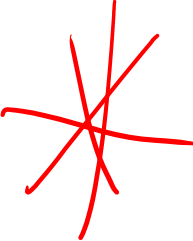
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$$\rho = \frac{\gamma p}{a^2}$$

Area-Mach Number Relationships

$$\frac{dp}{dA} = \left(\frac{\gamma M^2}{1 - M^2} \right) \frac{p}{A}$$

$$\frac{d\rho}{dA} = \left(\frac{M^2}{1 - M^2} \right) \frac{\rho}{A}$$


$$\frac{dV}{dA} = - \left(\frac{1}{1 - M^2} \right) \frac{V}{A}$$

Area-Mach Number Relationships

$$\frac{dV}{dA} = - \left(\frac{1}{1 - M^2} \right) \left(\frac{V}{A} \right)$$

Positive *Positive*

- For a subsonic flow ($M < 1$)

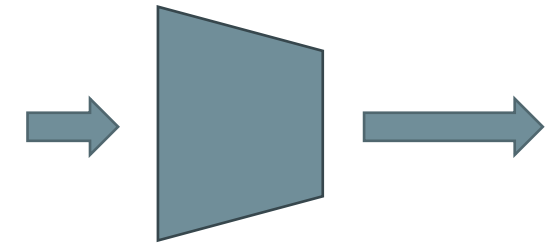
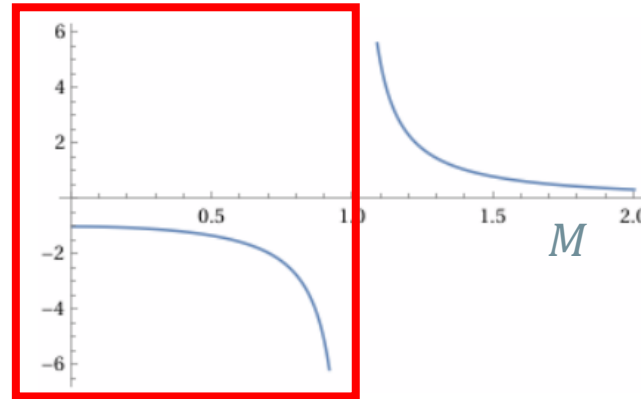
- $\frac{dV}{dA} < 0$ (from plot)

$$\frac{dV}{dA} = \frac{dV}{dx} \frac{dx}{dA} < 0$$

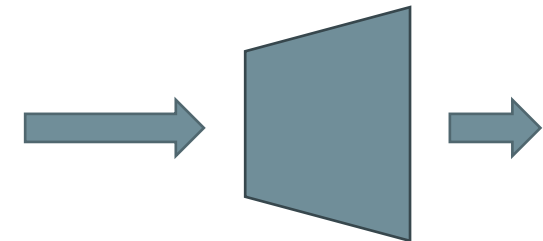
$$\text{If } \frac{dA}{dx} < 0 \quad \text{If } \frac{dA}{dx} > 0$$

$$\text{then } \frac{dV}{dx} > 0 \quad \text{then } \frac{dV}{dx} < 0$$

$$- \left(\frac{1}{1 - M^2} \right)$$



Subsonic Nozzle



Subsonic Diffuser

Area-Mach Number Relationships

$$\frac{dV}{dA} = - \left(\frac{1}{1 - M^2} \right) \left(\frac{V}{A} \right)^2$$

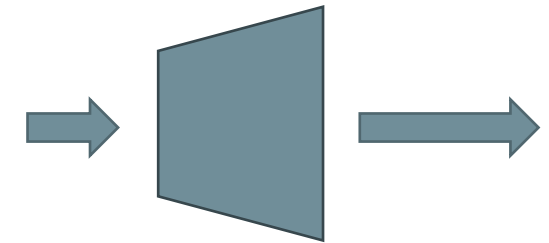
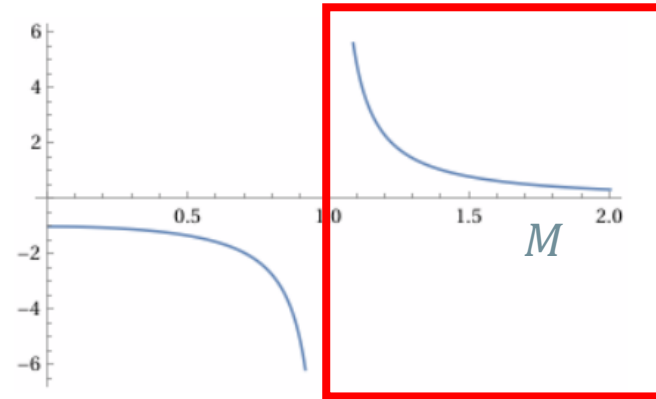
- For a supersonic flow ($M > 1$)

- $\frac{dV}{dA} > 0$

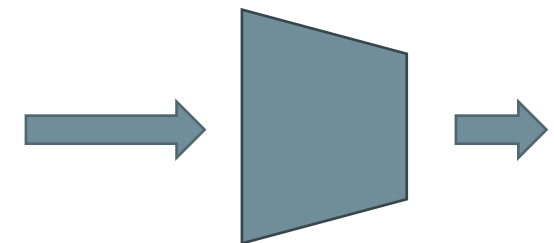
$$\frac{dV}{dA} = \frac{dV}{dX} \frac{dX}{dA} > 0$$

$$\begin{array}{l} \text{If } \frac{dA}{dX} > 0 \\ \text{then } \frac{dV}{dX} > 0 \end{array} \quad \begin{array}{l} \text{If } \frac{dA}{dX} < 0 \\ \text{then } \frac{dV}{dX} < 0 \end{array}$$

$$- \left(\frac{1}{1 - M^2} \right)$$



Supersonic Nozzle



Supersonic Diffuser

Area-Mach Number Relationships

$$\frac{dp}{dA} = \left(\frac{\gamma M^2}{1 - M^2} \right) \frac{p}{A}$$

$$\frac{d\rho}{dA} = \left(\frac{M^2}{1 - M^2} \right) \frac{\rho}{A}$$

$$\frac{dV}{dA} = - \left(\frac{1}{1 - M^2} \right) \frac{V}{A}$$

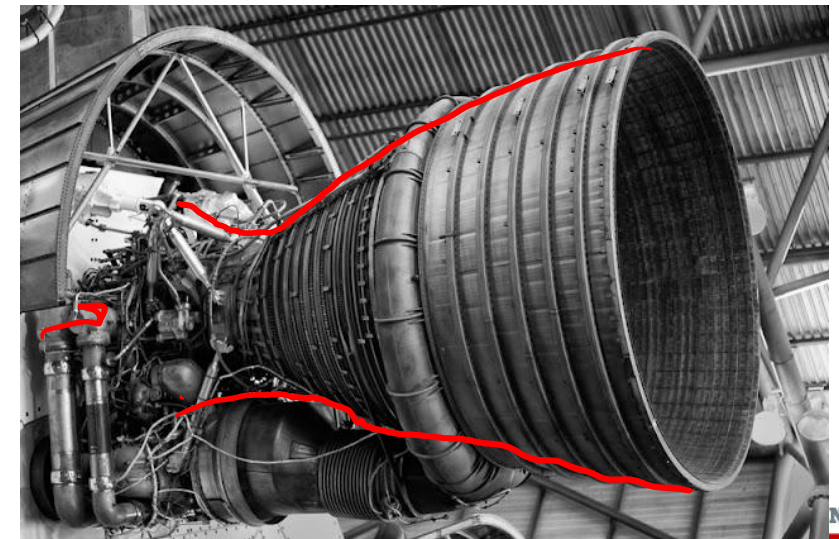
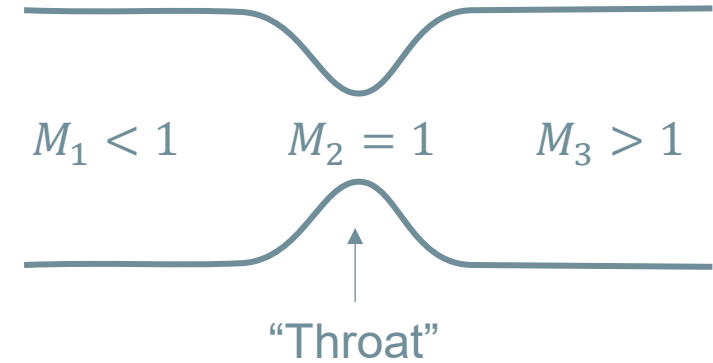
As the area decreases:

	Subsonic (M<1)	Supersonic (M>1)
Pressure (p)	Decreases	Increases
Density (ρ)	Decreases	Increases
Velocity (V)	Increases	Decreases

Can we ever accelerate a subsonic flow to $M > 1$?

Converging-Diverging Nozzles

- DeLaval nozzle
 - Converging nozzle until $M = 1$
 - Diverging nozzle afterwards
 - **Must reach $M = 1$ in throat**
 - Otherwise, $M_3 < 1$
- This requires specific design
 - Compute nozzle in two parts
 - $1 \rightarrow 2$ is the first (subsonic) process
 - $2 \rightarrow 3$ is the second (supersonic) process
 - We looked at a converging subsonic nozzle
 - Follow same principles, but set $M_2 = 1$



Converging-Diverging Nozzles

- DeLaval nozzle

- Example: Assume Air, $M_1 = 0.5$, $A_1 = 1 \text{ m}^2$, $p_1 = 1 \text{ MPa}$, $M_3 = 1.5$

$$A_2 = A_1 \frac{h_1}{h_2} \left(\frac{1 + \frac{\gamma-1}{2} M_1^2}{1 + \frac{\gamma-1}{2} M_2^2} \right)^{\frac{\gamma+1}{2(\gamma-1)}}$$

$$= (1 \text{ m}^2) \left(\frac{0.5}{1} \right) \left(\frac{1 + \frac{1.4-1}{2} (1^2)}{1 + \frac{1.4-1}{2} (0.5^2)} \right)^{\frac{1.4+1}{2(1.4-1)}} = 0.746 \text{ m}^2$$

$$A_3 = A_2 \frac{M_2}{M_3} \left(\frac{1 + \frac{\gamma-1}{2} M_3^2}{1 + \frac{\gamma-1}{2} M_2^2} \right)^{\frac{\gamma+1}{2(\gamma-1)}}$$

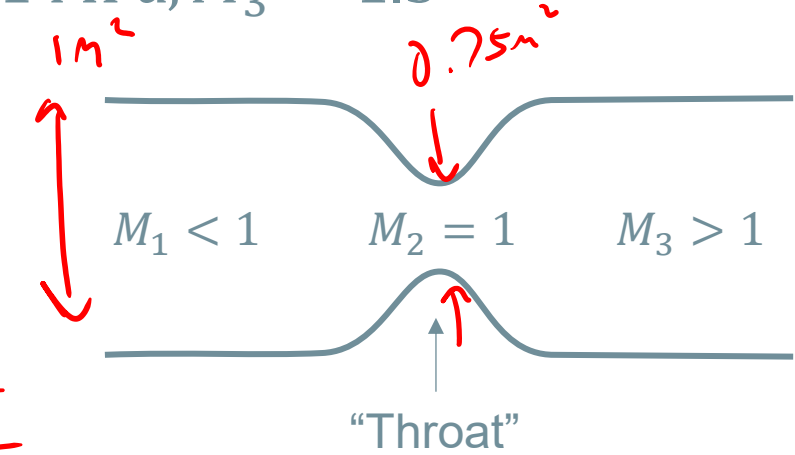
$$A_3 = 0.8777 \text{ m}^2$$

$$p_2 = p_1 \left(\frac{1 + \frac{\gamma-1}{2} M_1^2}{1 + \frac{\gamma-1}{2} M_2^2} \right)^{\frac{\gamma}{\gamma-1}}$$

$$p_2 = 0.954 \text{ MPa}$$

$$p_3 = p_2 \left(\frac{1 + \frac{\gamma-1}{2} M_2^2}{1 + \frac{\gamma-1}{2} M_3^2} \right)^{\frac{\gamma}{\gamma-1}}$$

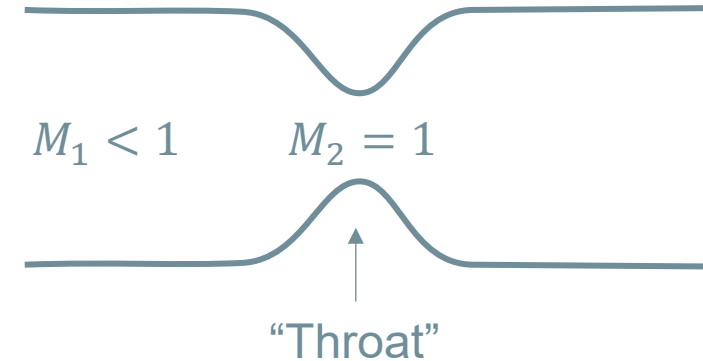
$$p_3 = 0.992 \text{ MPa}$$



- Pressure plays a critical role in controlling this flow

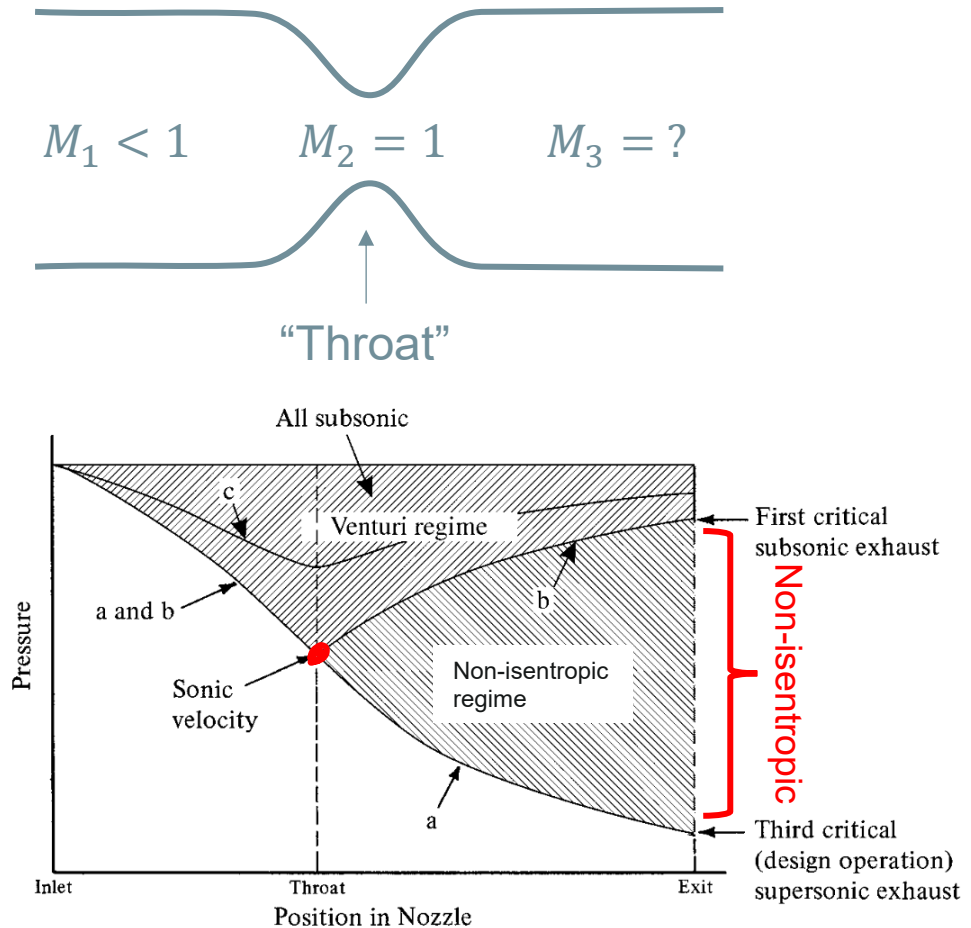
Converging-Diverging Nozzles

- Choked Flow
 - Cannot have $M_2 > 1$ since area decreases from 1 to 2
 - Flow must be at $M_2 = 1$ in the throat (otherwise $M_3 < 1$)
 - Max nozzle flow rate $\dot{m}_{\max} = \rho_2 U_2 A_2$
 - This condition is called “choked flow”



Converging-Diverging Nozzles

- If the throat is subsonic, the diverging section will be as well
 - Venturi regime (e.g., c)
- If the throat is at $M_2 = 1$, the diverging section can be subsonic or supersonic depending on exit pressure
 - First critical - subsonic exhaust (b)
 - Third critical - supersonic exhaust (a)
- Discuss everything between later



Friction Losses in Nozzles

- Only isentropic nozzles and diffusers so far – how do we account for losses?

Note: There are many ways to define these efficiencies based on different quantities. These are the ones we'll use in this course

- Isentropic efficiency of a nozzle:

$$\eta_{\eta} = \frac{\Delta h_{actual}}{\Delta h_{ideal}} = \frac{h_{t1} - h_2}{h_{t1} - h_{2s}} = \frac{T_{t1} - T_2}{T_{t1} - T_{2s}}$$

$\checkmark \frac{V_1^2}{V_{2s}^2}$

- Isentropic efficiency of a diffuser:

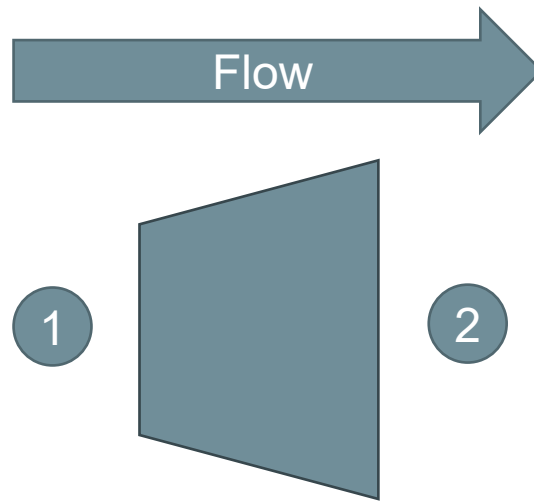
$$\eta_d = \frac{\Delta h_{actual}}{\Delta h_{ideal}} = \frac{h_{t2} - h_1}{h_{t2s} - h_1} = \frac{T_{t2} - T_1}{T_{t2s} - T_1}$$

This form is valid for ideal gases with constant specific heat, which is a decent approximation when working with relatively small temperature increases (i.e., not combustion)

Friction Losses in Nozzles

- Example: Consider a non-isentropic nozzle with constant c_p

- $M_1 = 3$
- $T_1 = 600 \text{ K}$
- $\gamma = 1.4$
- $M_2 = 6$
- $\eta_n = 0.85$



$$T_{+1} = T_1 \left(1 + \frac{\gamma-1}{2} M_1^2 \right)$$

$$= (600 \text{ K}) \left(1 + \frac{1.4-1}{2} (3^2) \right) = 1680 \text{ K}$$

For isentropic case, $T_{2s} = T_{+1}$

$$T_{2s} = T_{+2s} \left(1 + \frac{\gamma-1}{2} M_2^2 \right)^{-1}$$

$$T_{2s} = (1680 \text{ K}) \left(1 + \frac{1.4-1}{2} (6^2) \right)^{-1}$$

$$T_{2s} = 204.81 \text{ K}$$

- Find the **static** outlet temperature T_2

$$\eta_n = \frac{T_{+1} - T_2}{T_{+1} - T_{2s}} = 0.85 = \frac{1680 \text{ K} - T_2}{1680 \text{ K} - 204.81 \text{ K}} \rightarrow T_2 = 426.15 \text{ K}$$